Hritamm sarkar

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EDUCATION

Narula Institute of Technology 2025 – 2029 [Ongoing]
B.Tech in Computer Science and Engineering – Electronics and communication *GPA 8.03/10*
Techno India Group Public School 2025
Higher Secondary – Class XII *66%*
Jalpaiguri Zilla School 2022
Secondary Education – Class X *65%*

TECHNICAL SKILLS

Languages: Python, Java, C, SQL
Backend: FastAPI, REST APIs, Spring Boot
Databases: PostgreSQL,
Frontend: HTML, CSS
Cloud & DevOps: Docker, GitHub Actions, Vercel, Linux
Embedded Systems: Microcontrollers,Arduino, ESP32
Concepts: Data Structures & Algorithms, Object-Oriented Programming, System Design, DBMS, Computer Networks, REST Architecture

PROJECTS

RetinaScan AI — Explainable Diabetic Retinopathy Screening
Python, PyTorch, EfficientNet, OpenCV, FastAPI, Grad-CAM, PostgreSQL

- Developing an **AI-assisted diabetic retinopathy screening system** for analyzing retinal fundus images and classifying disease severity.
- Developing an **EfficientNet-based deep learning pipeline** with image preprocessing and quality assessment for reliable retinal image analysis.
- Implementing **Grad-CAM explainability and retinal lesion/vessel analysis** to provide visual evidence alongside predictions and support ophthalmologist referral decisions.

Automatic Sun Tracker — Arduino-Based Solar Tracking System
Arduino, LDR Sensors, Servo Motors, Embedded C/C++

- Built an automatic solar tracking system using Arduino and LDR sensors to detect the direction of maximum light intensity.
- Programmed servo motor control to automatically adjust the solar panel orientation toward the light source.
- Implemented sensor-based feedback logic for continuous tracking and optimized panel alignment.
- Gained hands-on experience in embedded programming, sensor interfacing, motor control, and basic control systems

BorderVision AI — AI-Powered Border Surveillance & Video Analytics
Python, FastAPI, OpenCV, YOLOv8, ByteTrack, WebSockets, Next.js, TypeScript, Leaflet

- Built a real-time multi-camera video analytics platform for AI-based object detection, tracking, geofencing, and incident monitoring.
- Developed a YOLOv8 + ByteTrack computer vision pipeline for object detection and multi-object tracking, with CLAHE-based enhancement for low-light CCTV footage.
- Implemented polygon geofencing and tripwire detection with real-time alerts using WebSockets, alongside a GIS dashboard for camera FOVs, incidents, and forensic snapshots.

EXPERIENCE

Johnson & Johnson MedTech — Robotics & Controls Job Simulation
***Virtual Experience* | *Forage* — July 2026**

- **Completed practical tasks focused on robotics and control systems in a medical technology context.**
- **Explored design and redesign considerations for surgical robotic systems.**
- **Analyzed engineering requirements and developed solutions for robotics-related challenges.**
- **Gained exposure to robotics, control systems, and engineering design processes.**